

Multilingual Plagiarism Detection Using GNNs and Syntax-Semantic Knowledge Graphs

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ABSTRACT Plagiarism has become an increasingly serious concern, especially within academia, driven by the unlimited availability of online information. It refers to the uncredited use of another author's text, images, code, graphics, or ideas. There are several types of plagiarism as direct copying, paraphrasing, reformulating, or even translating across languages. Historically, approaches for Cross Language Plagiarism Detection (CLPD) are centered on linguistic and structural cues, multilingual alignment processes, or a measurement of perceived similarity based on concepts to compare. In this study, we propose a new hybrid Approach for CLPD, which combines semantic information from WordNet with the syntactic structure from Universal Dependencies, then these relations are modeled in knowledge graphs for multiple language pairs. In detail, we utilize RotatE for relation embedding, GCN for node embedding, and GAT for improvement in learning contextual representations, to build and embed the graphs. To prove the efficiency of the new hybrid method, three approaches are compared: (1) using only WordNet to build the knowledge graph, (2) using only Universal Dependencies, and (3) the proposed hybrid approach. Our approach demonstrates clear improvements over state-of-the-art baselines across three language pairs: English-Spanish, English-French, English-Arabic, and English-German, on datasets including PAN11, JRC-Acquis, Europarl, Wikipedia, OPUS, and a curated set of conference papers. The proposed hybrid approach achieves Plagdet scores of 98.50% for English-Spanish, 89.02% for English-German, 77.65% for English-Arabic, and 77.67% for English-French.

INDEX TERMS Cross language plagiarism, knowledge graphs, WordNet, Universal Dependencies, RotatE, GNN

I. INTRODUCTION

Cross-Language Plagiarism (CLP) represents a significant and increasingly common form of academic misconduct. It occurs when an original text is translated from one language into another without providing appropriate attribution to the source. Detecting CLP can be particularly challenging in certain linguistic contexts for example, in regions such as China, where academic English is not widely mastered. Despite its importance, CL plagiarism can manifest in various ways, including copying and pasting without citation, translating content from other languages, or paraphrasing while altering writing style. It may also arise in cases of self-plagiarism or even the unauthorized reuse of original ideas, sometimes combining several of these forms. To address the growing challenges of CLPD, recent research has explored new techniques, including the use of knowledge-graph-based approaches. Already been

successfully applied in various fields such as information retrieval [1], medicine [2], cybersecurity [3], and recommendation systems [4]. This method allows representation of texts as semantic structures, no matter the language used. It allows for the detection of conceptual similarities between documents that are translated or reformulated, and it can detect instances of plagiarism from translation, paraphrasing, or idea theft, even when the form is linguistically different from one language to another. In this case, vector representations (or embeddings) of entities (or nodes) and relations for Knowledge Graphs (KG) are the means by which we will examine semantic similarity in multilingual texts. Unfortunately, most approaches that are developed today have really centered on semantic or lexical information without providing a representation of the syntactic structure that contributes to meaning interpretation. Universal Dependencies can provide syntactic information

and, when complemented by semantic relations in WordNet, allow for more comprehensive and linguistically rich representation of KG. Each of these aspects permits a deeper representation of the semantic and syntactic aspects of language, which results in improved representation across language relationships and detection of subtleties across texts. Utilizing these integrated representations, graph-based neural models provide some very effective ways to learn directly from the richer semantic-syntactic structure of KG. Graph Convolutional Networks (GCNs) enable learning node embeddings by incorporating graph structure and neighborhood information, proving useful in various domains such as bioinformatics [5], recommendation [6], and sentiment analysis [7]. In parallel, models like RotatE are employed to encode relations by modeling them as rotations in a complex vector space, effectively capturing translated or paraphrased semantic relations between entities and thus strengthening the robustness of CLPD systems [8]. In this paper, we present a new adversarial hybrid method called Cross-Lingual Universal Dependencies, WordNet, and Graph Neural Network (CL-UniDepWord-GNN) that employs semantic knowledge from WordNet and syntactic structures from Universal Dependencies. The proposed framework is designed for CLPD across four language pairs: English-Spanish (En-Es), English-French (En-Fr), English-Arabic (En-Ar), and English-German (En-De).

This study makes the following major contributions:

- A new hybrid approach CL-UniDepWord-GNN is proposed, combining a semantic knowledge from WordNet and syntactic information from Universal Dependencies. This method is applied for the first time in CLPD within a knowledge graph framework that supports multiple language pairs.
- The CL-UniDepWord-GNN model presents a novel multi-level embedding architecture that sequentially integrates different representation spaces: RotatE for relational embeddings, GCN for the structural representation of nodes, and GAT for improved context representation learning while constructing and encoding the graph.
- A comparative evaluation was conducted on three strategies based on WordNet exclusively, Universal Dependencies only, and combining both resources, which is the proposed CL-UniDepWord-GNN hybrid approach.
- The methodology was assessed across four language pairs, English-Spanish, En-Fr, En-Ar, and En-De, utilizing datasets including PAN11, JRC-Acquis, Europarl, Wikipedia, OPUS, and a curated set of conference papers, and evaluated by means of standardized metrics, such as Precision, Recall, Granularity, and Plagdet.

The rest of this paper is structured as follows. Related research on plagiarism detection methods using knowledge graphs and graph neural network models is reviewed in Section 2. Section 3 explains the proposed methodology in detail, whereas Section 4 describes the experimental setup and discusses the resulting outcomes. Lastly, Section 5 summarizes the paper's main conclusions and explores possible future study avenues.

II. RELATED WORK

There is still only a limited amount of research on CLPD that involves the usage of KG and employs various multilingual semantic networks like BabelNet and Wikidata, among others. This section presents studies that specifically make use of knowledge graphs in relation to CLPD.

A. Knowledge graph-based approaches for CLPD

They were among the first authors [9] to address the CLPD problem using KG, specifically leveraging the BabelNet resource. Their study focused on different language pairs, and the proposed model achieved a PlagDet (PlagDet is a standard metric used to evaluate plagiarism detection performance in PAN competitions) of 59.43% for the Spanish-English pair and 51.44% for the German-English pair. The authors [10] proposed cross-language knowledge graph analysis (CL-KGA) was applied to two cases: Paraphrase Translation (PT) and Automatic Translation (AT). The AT method achieved a PlagDet score of 60.87% for the Spanish-English language pair and 52.96% for the German-English pair, significantly outperforming the PT method, which yielded a PlagDet score of 9.93% for Spanish-English and 10.06% for German-English. In [11] proposed CL-PD Ontology-Based Similarity Analysis (CL-OSA) method utilizes deep semantic analysis to assess document similarity, leveraging entities and relationships extracted from the Wikidata open knowledge graph. In this approach, documents are categorized by topic, with each word annotated for part-of-speech (POS). Entities are identified and used to represent the documents as entity vectors derived from Wikidata. The relationships between entities are then quantified using cosine similarity, and candidate documents are ranked based on their similarity scores. The CL-OSA model achieved a PlagDet of 57.3% for the Spanish-English language pair and 52.1% for German-English, as evaluated on the PAN-PC-11 dataset. In [12] proposed a CLPD system that represents text fragments using knowledge graphs based on BabelNet. The system leverages Distributed Concept Weighting (DCW) and Word Sense Disambiguation (WSD). DCW helps express the strength of association between concepts (synsets) by generating distributed representations using BabelNet's synsets and the Skip-gram model to obtain vector representations of concept definitions. WSD addresses the challenge of determining the meaning of polysemous words, as they can have multiple interpretations. The combination of WSD and DCW, utilizing the Skip-gram model, yields significant results, achieving a PlagDet score of 66.3% for

the Spanish-English language pair and 59.5% for German-English. The authors of this paper [13] proposed a Knowledge-Based Similarity (KBSim) approach to compare documents written in different languages. This work introduces two main contributions: (1) a KG built using BabelNet, and (2) a new cross-lingual knowledge-based similarity measure (KBSim) that combines the KG with a traditional TF-IDF vector representation. The KBSim approach achieved an accuracy of 61.53% for the Spanish-English pair, 68.72% for German-English, 71.03% for French-English, and 54.87% for Italian-English.

B. GNN based approaches for representation learning

Beyond knowledge graph-based approaches for cross-language plagiarism detection, recent advances in graph neural networks have explored more general representation learning frameworks for modeling complex relational structures. Although these methods are not specifically designed for CLPD, they provide important insights into how graph-based embeddings and relational learning can be leveraged for structured data modeling.

Recently, several graph-based learning approaches have been proposed.

For instance, Cognitive Representation Optimization for Knowledge Tracing (CRO-KT) [16] models student learning processes using a bipartite graph composed of questions and skills, where representations are learned from student question interaction data via GCN-based embeddings. This approach does not explicitly capture heterogeneous semantic relations and also incorporates mechanisms to mitigate noise in student responses, such as guessing and slipping effects. Beyond standard GCN-based models, attention-based graph neural networks have also been explored for representation learning. In [15], proposed PatentGrapher, a hybrid model that combines Roberta (a pretrained language model) and GAT (Graph Attention Network), is designed to detect plagiarism in the full claim texts of Chinese patents. The dataset used consists of 10,317 authorized patents. The model applies a local-to-global strategy, where Roberta extracts local semantic features from each claim, while GAT captures the global relationships between claims within a graph structured according to internal references. The Roberta-GAT model achieves an accuracy of 91.64%, precision of 91.85%, recall of 91.57%, and an F1-score of 91.61%. More recently, Transformer-based architectures have been adapted for graph representation learning. In [14] proposed Cross-Lingual Graph Transformer-based Analysis (CL-GTA) approach employs knowledge graphs constructed using WordNet. It utilizes a specialized variant of Graph Neural Network (GNN) called a graph transformer, which leverages an attention mechanism to assign weights to all entities and semantic relations. To capture differences between nodes, a Bi-directional Gated Recurrent Unit (BiGRU) was implemented, while cosine similarity was used as the measure of similarity. The CL-GTA model delivered significant results, achieving a PlagDet of 62% for the Spanish-English language pair, 58.4% for French-English,

and 52.2% for Ar-En. In addition, the Graph Kernel Transformer (GraKerformer) [17] further extends Transformer-based graph learning by integrating shortest-path graph kernels (SPGK) into the attention mechanism. This allows the model to incorporate structural information more effectively, improving its ability to capture both local and global graph dependencies in unsupervised graph representation learning tasks.

Overall, CRO-KT focuses on interaction-based learning in bipartite educational graphs using GCNs, while PatentGrapher combines language models with GAT for attention-based reasoning. CL-GTA and GraKerformer extend these approaches toward Transformer-based models that better capture global structural information. However, most methods rely on interaction data, structural modeling, or general graph properties. Even CL-GTA, despite using WordNet, does not jointly exploit syntactic information, limiting multi-level linguistic representation. In contrast, our approach uses a knowledge graph enriched with semantic, syntactic, and cross-lingual relations, combined with GCN, RotatE, and GAT for richer multi-relational learning.

Table 1 presents a comparative analysis of techniques for CLPD. The main aspects are the following methods:

- **Multilingual Semantic Network (MSN):** specifies the type of semantic network used.
- **Methods and Techniques:** presents the methods (AT, PT, WSD, DCW) and techniques (Skip Gram, TF-IDF, GRU).
- **Similarity Metrics:** indicates the similarity measure used, such as Cosine, Euclidean Distance (ED), Jaccard, LCS, Dice Coefficient (DC), etc.
- **Language Pairs:** shows the couple of languages studied.
- **Dataset:** presents the different dataset used, such as the United Nations Parallel Corpora (UNPC) and King Saud University Corpus (KSUC).
- **Performance:** presents the performance metrics employed, including Plagdet (Pl), Precision (P), Accuracy (A), Recall (R), and Granularity (G).

Table 1. Comparison of CLPD approaches based on the knowledge graph.

Ref.	MSN	Methods & Techniques	Similarity	Language-pair	Dataset	Performance (%)
[9]	BabelNet		Conceptual	ES-EN	PAN-PC-11	Pl : 59.43, R : 51.83, P : 70.59, G : 1.00
[10]	BabelNet	AT PT	Conceptual	DE-EN ES-EN	PAN-PC-11	Pl : 51.44, R : 44.33, P : 63.19, G : 1.01 Pl : 60.87, R : 53.99, P : 70.36, G : 1.00
[11]	Wikidata		Cosine	DE-EN ES-EN DE-EN	PAN-PC-11	Pl : 52.96, R : 46.71, P : 63.06, G : 1.01 Pl : 57.3, P : 72.3, R : 47.4, G : 1.00 Pl : 52.1, P : 67.2, R : 42.5, G : 1.00
[14]	WordNet	GRU GNN	Cosine	ES-EN	PAN-PC-11 JRC-Acquis	Pl : 62, P : 63.06, R : 46.71, G : 1.01
				FR-EN	Europarl, Wikipedia	Pl : 58.4, P : 50.6, R : 69, G : 1.00
				AR-EN	Conference papers English-Arabic parallel corpora.	Pl : 52.2, P : 40.9, R : 57.6, G : 1.101
[12]	BabelNet	DCW, WSD Skipgram	Dice coefficient,	ES-EN	PAN-PC-10 PAN- PC-11	Pl : 66.3, P : 76.1, R : 58.8, G : 1.00
		SenVec	Cosine	DE-EN		Pl : 59.5, P : 70.3, R : 51.6, G : 1.00
[13]	BabelNet	TF-IDF	Dice coefficient,	EN-ES	Multilingual Reuters	A : 61.53
			Cosine	DE-EN	Collection	A : 68.72
				EN-FR		A : 71.03
				IT-EN		A : 54.87

Table 1 presents a comparative study of works based on KG and the methods integrated to improve the performance of graph representations. For the preprocessing stage, the most commonly used techniques are lemmatization and POS tagging. Additionally, for multilingual semantic networks, BabelNet is the most widely used resource. Various datasets are devoted to CLPD, with PAN-PC-11 emerging as the dominant benchmark, notably used in the international competition on plagiarism detection. Several language pairs have been investigated, including En-Es, En-De, En-Fr, and En-Ar. However, Arabic has not been extensively studied, despite the fact that language characteristics strongly influence model performance. For example, the English-Spanish pair achieved 66.3% PlagDet, while the English-German pair reached lower performance levels. To measure graph similarity, most studies rely on Cosine similarity and the Dice coefficient. The integration of DCW and WSD methods has enhanced the performance of CLPD based on knowledge graph analysis (CL-KGA), achieving a PlagDet of 66.3% for the Spanish-English pair. The use of GNN achieved a PlagDet of 62% for Spanish-English. Traditional vector representations, such as TF-IDF, yielded 68.72% accuracy for En-De and 61.53% for En-De. However, these state-of-the-art results remain limited, with performance not exceeding 68.3%, highlighting the need for more robust and linguistically grounded approaches.

These findings bring the following three aspects of the current approach to the fore as areas for improvement: insufficient exploitation of syntactic features, poor treatment of cross-lingual connections, and poor performance on morphologically complex languages.

Likewise, our proposed solution specifically targets the aforementioned limitations by presenting a conceptual framework that combines the semantic information found in WordNet with the syntax information parsed from the Universal Dependencies for incorporation into the KG. By strategically implementing the synergies offered by the different linguistic dimensions enabled by the KG, the plagiarism detection task will therefore be improved because of the enhanced modeling capabilities afforded by the more sophisticated architectures.

III. BACKGROUND CONCEPTS

C. WORDNET

WordNet is a lexical database that describes the English language in a way that was developed by the Cognitive Science Laboratory at Princeton University [18]. It groups words into synonym sets or "synsets," which represent a particular concept or meaning. Synsets are connected to other synsets through a wide variety of semantic relations (e.g., hypernymy and hyponymy) to model conceptual relations of meaning between words [19]. WordNet is widely used in Natural Language Processing (NLP) tasks such as word sense disambiguation and information retrieval. The concept of Multilingual WordNet emerged as an extension of the original Princeton WordNet in order to cover multiple languages. The first large-scale initiative was EuroWordNet, which initially supports 4 languages: Dutch, Italian, Spanish, and English [20]. In an extension to the project, the database is extended with French, Estonian, and Czech [21]. Later, similar projects were developed for other regions, including BalkaNet for Balkan languages [22] and IndoWordNet for Indian languages [23]. For languages not included in such

collaborative projects, independent resources were created, for example, GermaNet for German [24] and FinnWordNet for Finnish [25]. This chart provides an estimate of the proportion of synsets available for four languages (English, Spanish, French, and German) in WordNet. The values are approximate, as the exact number depends on the specific WordNet version and the linguistic resource used. The data highlight the dominance of English, which accounts for around 59% of the synsets, while the other languages remain comparatively less covered [26].

D. UNIVERSAL DEPENDENCIES

The Universal Dependencies (UD) is a cross-linguistically consistent grammatical annotation system for NLP. The goal of the framework is to create a common standard across many languages to represent syntactic structure and morphological features. To do this, the framework enables multilingual parsing, linguistic comparison and cross-language transfer utilizing equal meaning across languages through the definition of all languages using the same set of syntactic relations and forms of speech, whilst allowing for language-specific variations and subtleties [27], [28]. UD follows a dependency grammar approach, in which syntactic structure is represented as a set of binary relations (dependencies) between words in a sentence, with a single root and directed arcs connecting heads to dependents.

UD has developed a comprehensive system of syntactic labels (nsubj, obj, obl, advmod) and POS tags (NOUN, VERB, ADJ), allowing for international use. Different languages have distinct grammatical structures, verb forms, and placements. To accommodate these differences, each language (sometimes each treebank within a language) has its own description of how to modify universal principles to comply with the characteristics of that language. Additionally, every language has treebanks labelled by either a native speaker or a trained linguist, ensuring that syntactic representations are accurate and true to the language in question. Additionally, morphological features such as tense, number, and case are described using a universal inventory of features, but are applied in ways that reflect the grammatical system of the target language [29], [30].

E. KNOWLEDGE GRAPH

A knowledge graph is a graphical representation of a knowledge base in which nodes represent entities, whether real (such as people, places, or objects) or abstract (such as concepts or events). The labeled edges capture the semantic relationships connecting these entities, thereby providing an explicit structure. This organization is typically expressed through (head-relation-tail) triplets, which form the fundamental unit of the graph [31].

F. GRAPH NEURAL NETWORKS

GNNs are a family of neural networks that have been developed to apply deep learning methods to graph-structured data, allowing for the aggregation and propagation of information, along with their neighborhood structure. Because GNNs' structure is based on the message-passing framework, at each layer, the node representation is updated based on its own features and its neighbors' features through learnable aggregation functions and transformation functions. This layered structure allows GNNs to capture increasingly complex patterns across local and global graph structures [32], [33].

Recent advances have further extended GNN capabilities through self-supervised and contrastive learning paradigms. In particular, multi-topology contrastive graph representation learning has been proposed to capture multi-granularity information across different graph structures, enabling richer and more expressive node representations by leveraging complementary structural views [34], [35].

Beyond these core tasks, GNNs have been used for diverse application domains, including bioinformatics for protein interface prediction [36], social networks [37], fraud detection [38], and recommender systems [39].

In building on these fundamentals, our approach exploits WordNet and Universal Dependencies within an integrated KG, combined with Graph Neural Networks that provide semantic information along with syntactic structure. It directly targets the current state-of-the-art limitations mentioned above, where most of the cross-lingual plagiarism detection methods are basically limited by relying on semantic resources alone, using superficial similarity measures, having limited syntactic modeling, and using partial semantic relations, which diminishes their performance, especially for morphologically rich languages. Our method uses the integrated KG of WordNet and Universal Dependencies, capturing both semantic equivalences and syntactic variations, while Graph Neural Networks learn deep, context-aware representations from its structure and thus improve performance and generalization across multiple language pairs, including morphologically rich languages like English-Arabic.

IV. METHODOLOGY

This section describes the methodological steps of the proposed approach. The first step is a general preprocessing step applied consistently across the dataset. The second step in this approach is to build a KG that gathers semantic relations using WordNet and syntactic relations with Universal Dependencies. The two layers of information are then combined into a single KG. The third stage involves graph integration, where relations are embedded with RotatE and nodes are represented with a GCN. Finally, for the classification task, we argue that a GAT is then used to fully leverage the enriched graph representations. Figure 1 shows the different stages in the methodology.

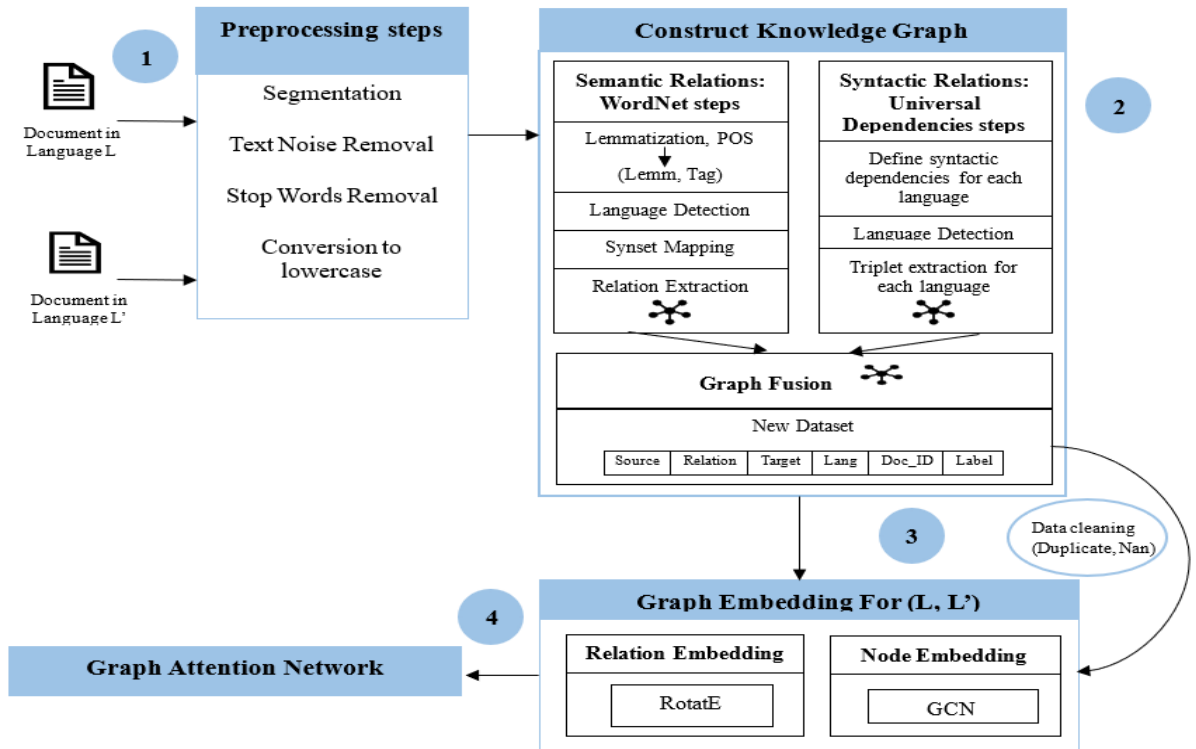


Figure 1. Proposed Methodology

A. DATASET

For the plagiarism detection experiments in this study, five different corpora were utilized: PAN11 [40], JRC-Acquis [41], Europarl [42], Wikipedia [43], OPUS [44], and a collection of Conference Papers [45]. The study targeted four bilingual settings: En-Fr, En-Es, En-Ar, and En-De. English was chosen as the pivot language, largely due to the nature of the available resources. For instance, PAN11, which is designed for the En-Es and En-De scenarios, consistently treats English as the main source language.

Regarding dataset composition:

- For En-Fr, the corpus contained 10,620 plagiarized pairs, drawn from Conference Papers and JRC-Acquis, alongside 9,442 non-plagiarized pairs sourced from Europarl and Wikipedia.
- For En-Es, we gathered 10,210 plagiarized pairs (from PAN11 and JRC-Acquis) and 9,423 non-plagiarized pairs (from Europarl and Wikipedia).
- For En-De, the dataset was built with 12,221 plagiarized pairs (from PAN11 and JRC-Acquis) and 9,999 non-plagiarized pairs (from Europarl and Wikipedia).
- For En-Ar, the dataset included 151,136 parallel sentence pairs taken from the Wikipedia component of the OPUS corpus, administered by the University

of Helsinki, and then balanced by creating another 151,136 non-parallel pairs through random sampling for a total of 302,272.

Table 2 describes the characteristics of the dataset used.

Table 2. Summary of Corpus Sources and Sizes for CLPD.

Language Pairs	Datasets	Pairs of Documents	Target	Total Number of Documents
English-French	Conference Papers	620	P	20,062
	JRC-Acquis	10,000	P	
	Wikipedia+Europarl	9,442	NP	
English-German	JRC-Acquis	9,999	P	22,220
	PAN11	2,222	P	
	Wikipedia+Europarl	9,999	NP	
English-Spanish	PAN11	210	P	19,633
	JRC-Acquis	10,000	P	
	Wikipedia+Europarl	9,423	NP	
English-Arabic	OPUS: Wikipedia	151,136	P	302,272
	Randomly generated	151,136	NP	

B. PREPROCESSING STEP

The first preprocessing step gets the documents in both languages ready for their transformation into graphs through

normalization. This process begins with the segmentation of sentences, which breaks the text into coherent and manageable segments of the text. The next step in the process is cleaning the text to remove noise, such as special characters and duplicate spaces, and removing stop words that contain no significant information. Finally, the text is lowercased, so consistency is maintained, thus facilitating the next steps of processing and representation.

C. KNOWLEDGE GRAPH CONSTRUCTION

A Knowledge Graph was developed using a hybrid approach that combines semantic and syntactic information. The semantic layer is obtained from WordNet, while the syntactic layer is from Universal Dependencies, which models grammatical structures in a language-independent context. Merging these two layers yields a graph that represents both the meaning and grammatical structure of the text. The detail steps are described in the following sections.

1) SEMANTIC RELATIONS EXTRACTION USING WORDNET

After we pre-processed the data, the next step was lemmatization and POS tagging so that we could represent each of the individual words as a combination of a (lemma, tag) pair. These pairs are treated in accordance with their language, the language having been identified from the original annotation of the raw data, and passed to the program via the lang parameter (i.e. Eng for English and Spa for Spanish). Once we have identified a (lemma, tag) pair for each word in a document, we make a call to WordNet using the lemma and its POS to determine the synsets for that word. After this, we grouped the words that originated from the same document together and then created a potential connection between those words by using a sliding window; each word is only compared to the immediate words next to it in a five-word span, thereby capturing local semantic relationships while preventing the exponential growth of computation through processing the entire text. For each (lemma, tag) pair which we have processed into the sliding window, we compare the synsets for each of the two words and create an edge in the graph linking the two terms if there is a semantic relationship (hypernymy, hyponymy, meronymy, holonymy, synonymy, antonymy, etc.) found in WordNet. The graph edge created is annotated with the relation type, document ID, language, and label associated with it. To support multiple languages, we relied on Open Multilingual WordNet, which extends the original English WordNet to several languages. For German, we specifically used OdeNet [46], a dedicated lexical database aligned with Princeton WordNet, ensuring more accurate coverage of German vocabulary and relations, while for Arabic, we employed the Arabic WordNet (AWN), a lexical-semantic resource developed as part of the Global WordNet project to provide comprehensive coverage of Arabic concepts and relations.

The semantic representation of the sentence "Ahmed reads a book" is shown in Figure 2 using WordNet. Each word is

linked to a synset (a lexical sense), and we see the words linked together based on semantic relations such as synonymy, hypernymy, or meronymy between the words. For example, we can see read is linked to book based on conceptual relations between them that reflect their lexical closeness and their position in WordNet hierarchy. This graph clearly shows the strengths and weaknesses of relying solely on lexical extraction. On one hand, we have WordNet making conceptual links between content words (i.e. read ↔ book), and on the other hand, they are not addressing named entities or syntactic roles (i.e., Ahmed as the agent). Thus, while the semantic graph gives us an overview of lexical meaning, it does not reflect the propositional structure of the original sentence (who does what). In our implementation, we limited the number of relations extracted per word to 3 in order to avoid representing a congested-looking graph and to maintain readability, but most importantly, while preserving meaningful semantic coverage. Three relations was found adequate to use and extract to scale to large and medium datasets since it balances paying attention to the richness of the semantic connections to speed of the extraction process. In this way, WordNet offers a dense but incomplete semantic basis for constructing meaning for sentences. To build a complete KG of sentence meaning, they should be combined with the syntactic graphs produced by UD, which specify grammatical roles such as subject and object. The relationship between meaning and syntax provides the full representation of meaning at the level of the sentence, with lexical relations and grammatical roles completing each other in this KG.

To formalize the construction of the semantic graph from WordNet, we define:

$$D = \{w_1, w_2, \dots, w_n\} \quad (1)$$

(1): the set of words in a document, where each word is represented by a tuple $(lemma_i, POS_i)$.

For each word w_i , we extract its set of synsets S_i using WordNet:

$$S_i = \text{WordNet}(lemma_i, POS_i) \quad (2)$$

Candidate word pairs for edge creation are defined using a sliding window of size $k = 5$:

$$C = \{(w_i, w_j) \mid |i - j| \leq k, i \neq j\} \quad (3)$$

An edge is created between two words w_i and w_j if at least one synset of w_i is semantically related to a synset of w_j in WordNet via a relation $r \in R$, where:

$R = \{\text{synonymy, hypernymy, hyponymy, meronymy, holonymy, antonymy}\}$.

$$e_{ij} = (w_i, w_j, r) \text{ if } \exists s_p \in S_i, s_q \in S_j \text{ such that relation}(s_p, s_q) = r \quad (4)$$

To prevent the graph from becoming overly dense, we limit the number of relations per word to 3:

$$|\{e_{ij} : w_i \text{ is fixed}\}| \leq 3 \quad (5)$$

The final semantic graph is defined as:

$$G_s = (V, E_s) \quad (6)$$

where:

$V = D$ and $E_s = \{e_{ij} \mid e_{ij} \text{ satisfies the above conditions}\}$.

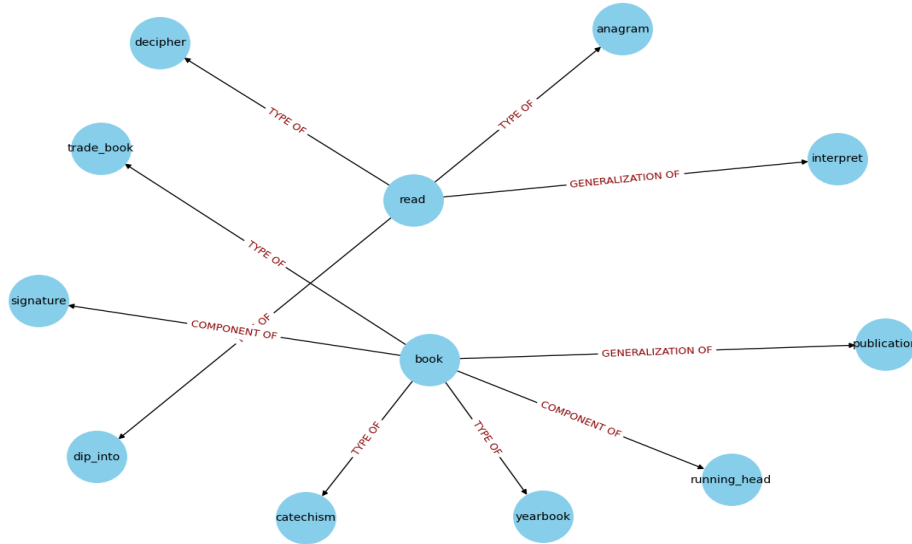


Figure 2. Example of a semantic graph generated by WordNet.

2) SYNTACTIC RELATIONS EXTRACTION USING UNIVERSAL DEPENDENCIES

In our approach, we leveraged UD to automatically retrieve syntactic relations as triplets (subject, verb, object) from text produced in a language L and a language L' . After applying spaCy linguistic models for each language (English, Spanish, German, and French), as well as the Stanford Stanza library prepared for Arabic, we identified grammatical dependencies such as *nsubj*, *dobj*, *obj*, *iobj*, and *pobj*, which we then converted into edges between terms, with metadata (document ID and label). This action allows for a sentence to be conceptualized as a syntactic graph, indicating relations between linguistic entities. The methodological strategy assumes that while the UD framework aims at standardizing relations, various specificities exist within the cross-language applications of UD (i.e., some relations seem to prevail and thus are standardized across languages, while others seem unique to endemic relationships across languages, i.e., German and its morphosyntactic relations, which follow from its richer grammatical structure). As such, our extraction of relations reflects the diversity of relationships involved in their grammatical properties and creates both a consistent foundation of substance across relations and a relational diversity between the languages in the corpus.

Figure 3 shows the syntactic dependency graphs derived from the same simple example in four languages: English, Spanish, German, and Arabic. The sentence used, "Ahmed reads a book," is simple, but the final dependence structures reveal a great deal of syntactic variation in and across languages. The graphs are generated from the same basis of representation in the UD framework, which represents syntactic dependency while continuing to represent

grammatical features of each system. Further, relation types like 'nsubj', 'obj', or 'det' do not always have equivalent direct relation types from language to language. Some remain quite stable, like the *nsubj* relation in English and Spanish; others vary due to varied morphosyntactic density. The basic issue is complicated further due to the complexity of constructing multilingual syntax graphs. Compound or complex sentences make the challenge even greater, as the extraction of nodes and relations becomes a problematic issue when the components are nested or span several clauses. Language-specific syntactic rules must be established to identify syntactic entities (nodes) and syntactic relationships (edges). Again, these rules rely on dependency relation data that are a feature of the grammar, and ultimately, extraction needs to be consistent from decomposition to decomposition while still being compatible across languages.

To construct this syntactic graph, we explicitly model the grammatical relationships between words by leveraging dependency structures obtained from Universal Dependencies, where each relation encodes a specific syntactic function.

Let D be the set of words in a document, as defined in Eq. (1), where each word represents a potential node in the syntactic graph.

After syntactic parsing using Universal Dependencies (UD), each sentence is represented as a set of dependency triplets:

$$T = \{(w_i, w_j, r)\} \quad (7)$$

where:

- $w_i, w_j \in D$ are words in the sentence,
- $r \in R_{UD}$ denotes the grammatical dependency type (e.g., *nsubj*, *dobj*, *obj*, *iobj*, *pobj*, etc.).

Each triplet thus corresponds to a directed edge in the syntactic graph, linking word w_i to word w_j with grammatical role r .

Each edge encodes both the structural dependency and its linguistic function, enabling the model to capture syntactic relationships between words.

The syntactic graph is then defined as:

$$G_t = (V, E_t) \quad (8)$$

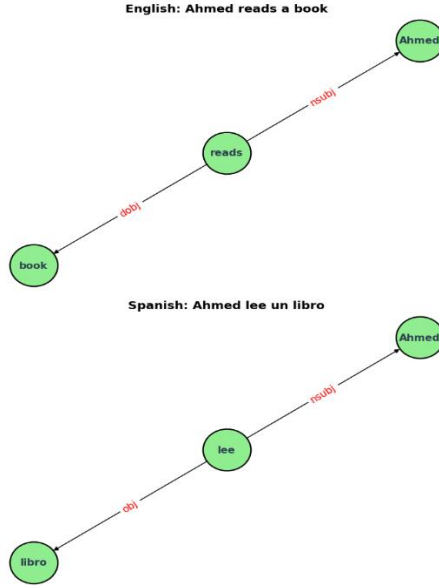


Figure 3. Example for Syntactic Dependency Graphs across Languages.

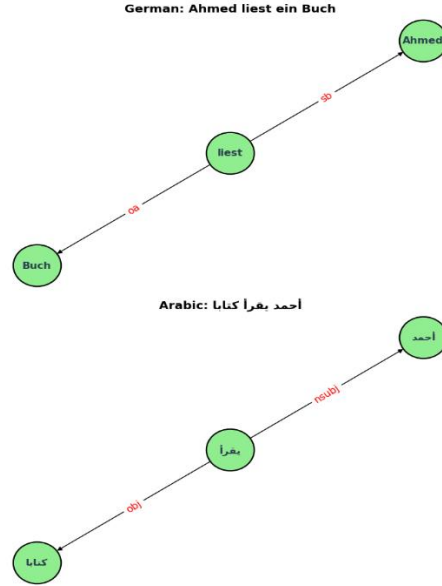
3) FUSION OF SEMANTIC AND SYNTACTIC INFORMATION

To create a fuller and more representative KG, the semantic relations obtained from WordNet were combined with the syntactic relations provided by UD. WordNet has good lexical-semantic information in the form of synonymy, hypernymy, hyponymy, meronymy, and antonymy that represent conceptual relationships between words. The UD source was used to provide syntactic dependencies (nsubj, dobj, obj, iobj, and pobj) that reveal how words relate to each other in sentence structures. By combining the two, we can say that the lexical-semantic KG will contain both 1) semantic relatedness (i.e., conceptual distance between words), and 2) syntactic structure (i.e., grammatical relationships to words in the sentences). The end product is a hybrid graph that is based on solid linguistics and stays true to being conceptually semantic while developing a richer representation for CLPD.

The hybrid syntax-semantic graph for the sentence “Ahmed reads a book” is displayed in Figure 4. In this figure, the syntactic dependencies produced using UD (i.e., nsubj between Ahmed and read and npadvmod between read and book) are fused with semantic relations derived from WordNet (e.g., TYPE OF, GENERALIZATION OF, COMPONENT OF). Therefore, the nodes read and book serve as anchors where both syntactic and semantic layers converge. This merger enables the graph to represent complementary linguistic dimensions: UD organizes the

where:

- $V = D$: the set of words in the sentence (nodes),
- $E_t = \{(w_i, w_j, r) \mid (w_i, w_j, r) \in T\}$: the set of syntactic edges annotated with relation type, document identifier, and label.



sentence based on grammatical functions, establishing Ahmed as the agent and book as the object of the action; whereas WordNet expands the same graph with conceptual links connecting read and book to broader or related lexical concepts (e.g., publication, interpret, yearbook). The benefit of having a single graph containing both representations is a more nuanced understanding of how entities (e.g., Ahmed, book) and concepts interact, which is a critical step toward CLPD, where it is essential to capture syntactic equivalence (i.e., the sentence's structure) along with semantic similarity (i.e., the concept represented).

To formally define this hybrid representation, we now introduce the construction of the fused syntax-semantic graph.

Let:

- $G_s = (V, E_s)$ be the semantic graph derived from WordNet.
- $G_t = (V, E_t)$ be the syntactic graph derived from UD.

The fusion of these two graphs results in a hybrid graph G_h defined as:

$$G_h = (V, E_h), \text{ with } E_h = E_s \cup E_t \quad (9)$$

where:

$$E_h = \{(w_i, w_j, r) \mid (w_i, w_j, r) \in E_s \text{ or } (w_i, w_j, r) \in E_t\} \quad (10)$$

The edges E_h integrate both semantic and syntactic relations, each annotated with their corresponding type r derived from either WordNet or UD. As a result, words such as “read” and “book” serve as anchor points where grammatical UD and conceptual WordNet relations converge.

The hybrid graph can be reformulated in a compact form as:

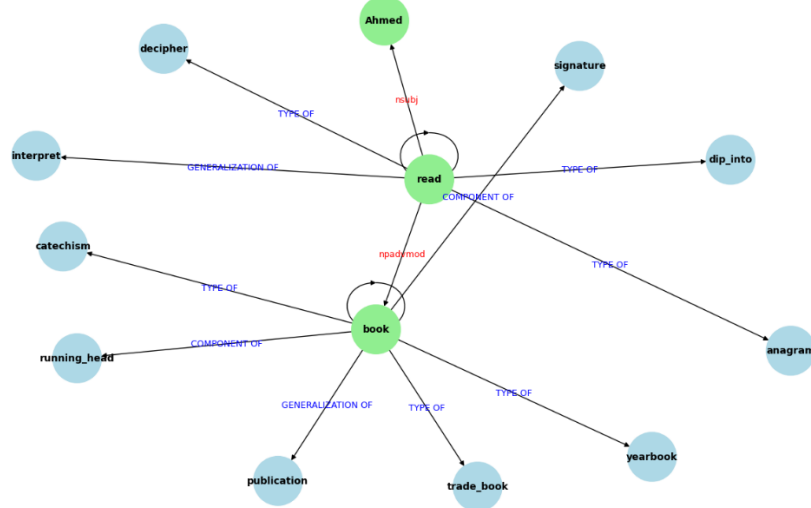


Figure 4. Example for Semantic-Syntactic Graph.

D. EMBEDDING GRAPH

1) NODE EMBEDDING

A Graph Convolutional Network (GCN) is a particular class of neural networks developed for processing graph-structured data directly. It extends the classical convolution operation originally developed for grid-like data such as images in CNNs, to graph domains, which are inherently irregular and non-Euclidean [47], [48]. This extension allows GCNs to extract meaningful representations from both node attributes and the structure of their relationships. While CNNs apply convolutional filters over regular, fixed-size neighborhoods in images, GCNs apply analogous filters over local neighborhoods of nodes in a graph, where the number and arrangement of neighbors may vary. These filters aggregate and combine feature information from neighboring nodes, capturing both structural and semantic patterns. The extracted local features are then propagated across multiple GCN layers, enabling the network to learn increasingly abstract and high-level representations [49].

GCN is used to learn node representations. For this purpose, an embedding layer is defined to associate each node with a dense vector of fixed dimension. The embedding matrix is initialized using the Glorot uniform method, and the forward function returns the node vectors that serve as input to the GCN layers for learning the structural dependencies of the graph.

To initialize node representations, each node in the graph is mapped to a continuous vector space through an embedding layer.

$$G_h = \{(w_i, w_j, r) \mid (w_i, w_j, r) \in G_s \text{ or } (w_i, w_j, r) \in G_t\} \quad (11)$$

Thus, G_h provides a unified and enriched representation of the sentence or document, which is essential for CLPD, as it enables the model to jointly capture structural equivalence and semantic similarity.

$$h_i^{(0)} = \text{Embedding}(w_i) \quad (12)$$

where $h_i^{(0)}$ denotes the initial embedding of node w_i , learned through an embedding layer that maps each node to a dense vector representation.

2) RELATION EMBEDDING

One of the main techniques for KG embedding is Rotation-based Embedding (RotatE), which represents entities as complex-valued vectors, while relations are modeled as rotations in the complex space. This method is particularly well-suited for encoding relational structures, as it captures various relational properties such as symmetry, antisymmetry, inversion, and composition, thereby enabling effective reasoning and KG completion. RotatE models relations geometrically by learning transformations that rotate the head entity vector toward the tail entity vector in the complex space. [50].

RotatE describes relations within the graph by modeling each relation as a rotation in complex space, where embedding vectors have been initialized as angles ranging from 0 to 2π . This notation can represent both type and direction of links between entities, and therefore provides a more informative representation of relations.

To model the interactions between nodes, each relation in the graph is represented as a learnable embedding vector, enabling the model to encode different types of semantic and syntactic connections.

$$r_{ij} = \text{Embedding}(\text{relation}_{ij}) \quad (13)$$

where r_{ij} represents the embedding of the relation between nodes i and j , learned as a vector representation inspired by the RotatE model.

E. GRAPH NEURAL NETWORK

In this research, we construct a model based on the GAT to learn enriched node and relation representations in a unified graph learning framework. Starting with node embeddings generated by GCN encoders for the language (L/L'), the first step involves passing these vectors to a GAT layer, which applies attention mechanisms to weight neighboring node contributions. We subsequently account for relations using the RotatE module, where each relation is modeled as a rotation in the embedding space. Relation embeddings are initialized as learnable parameters in a continuous space and jointly optimized during training, enabling task-specific relational modeling in an end-to-end manner.

The integration between GAT-derived node embeddings and RotatE relation embeddings is performed via concatenation, ensuring that structural and relational information are preserved without interference. Once the source node, relation, and target node embeddings are concatenated, the resulting representation is regularized using dropout to reduce overfitting and improve generalization on graph-structured data. Finally, an MLP (a stack of linear layers with ReLU and Dropout) projects the information into a decision space with a sigmoid activation for binary classification (0/1), indicating the presence or absence of a relevant link (e.g., plagiarism or not).

To capture contextual dependencies between nodes, the GAT updates node representations by aggregating information from their neighbors using attention mechanisms:

$$h'_i = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} W h_j \right) \quad (14)$$

where h'_i is the updated node representation, $\mathcal{N}(i)$ denotes the neighborhood of node i , α_{ij} are attention coefficients, and W is a learnable weight matrix.

After obtaining node and relation embeddings, the model integrates them into a unified representation for each pair of nodes.

$$z_{ij} = [h_i^{GAT} \parallel r_{ij} \parallel h_j^{GAT}] \quad (15)$$

where z_{ij} denotes the combined representation obtained by concatenating the source node embedding, the relation embedding, and the target node embedding.

Finally, the combined representation is passed through a classification layer to predict the existence of a relationship between nodes.

$$\hat{y} = \sigma(\text{MLP}(z_{ij})) \quad (16)$$

where $\hat{y}$ represents the predicted probability of a link between nodes, obtained through a multilayer perceptron followed by a sigmoid activation.

The overall architecture is illustrated in Figure 5, which summarizes the full pipeline from node encoding to final prediction.

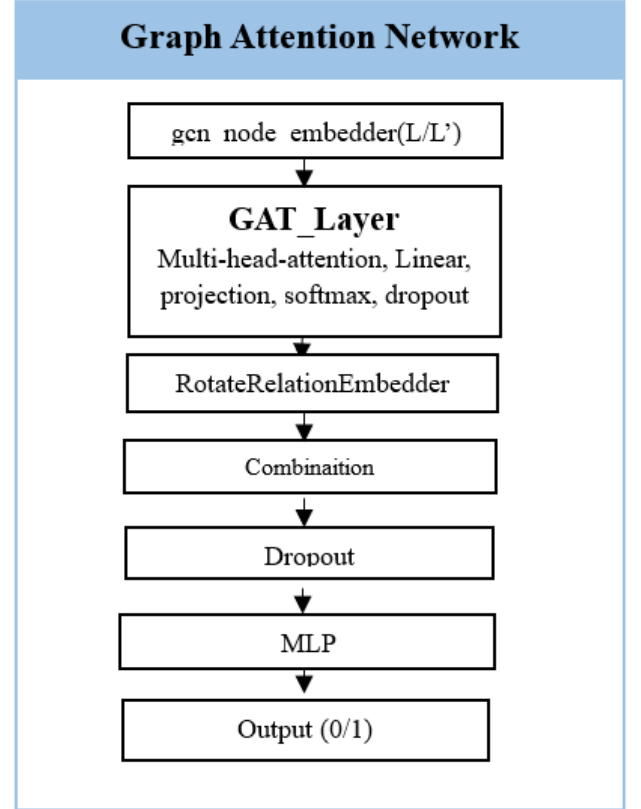


Figure 5. GAT Architecture.

F. Training Configuration

1) HYPERPARAMETER MODEL

In our approach, we used a GAT model and projected node representations into a 64-dimensional latent space. This dimensionality was chosen as a trade-off between representational capacity and computational efficiency, in line with previous work in graph representation learning. The GAT layer employs an attention mechanism with linear projection and dropout (0.3) to capture structural dependencies between nodes while reducing overfitting in dense graphs. The dropout rate was selected empirically to improve generalization performance. The parameters were optimized using the Adam optimizer with a learning rate of 0.001. Binary Cross-Entropy (BCELoss) was used as the loss function due to its suitability for binary classification tasks. We used a batch size of 32 to ensure stable gradient updates while maintaining computational efficiency. The model was trained for 10 epochs, which was sufficient to achieve convergence without overfitting, as confirmed through validation performance. GPU acceleration was used when available to reduce training time. Table 3 summarizes the hyperparameter configuration of the proposed model.

Table 3. Characteristics of GAT hyperparameters.

Hyperparameters	Value
embedding size	64
Dropout	0.3
Optimizer	Adam
Learning rate	0.001
Loss function	Binary Cross-Entropy (BCELoss)
Batch size	32
Number of epochs	10
Attention mechanism	Multi-head Attention

2) DECISION THRESHOLD SELECTION

Our research used a grid search for validation to create candidate thresholds (range 50-70%) in order to convert continuous outputs generated from the trained model into binary predictions needed for plagiarism detection for the purpose of identifying the optimal classification threshold. A variety of candidate thresholds, from 50 - 70%, were assessed using the validation dataset to allow for an equal balance between precision and recall. The threshold that produced the maximum PlagDet score was used because PlagDet provides more thorough evaluation by assessing precision, recall, and granularity together when determining whether a prediction is accurate. The final threshold value of 63% was fixed, and used evenly throughout all of the experiments with multiple languages, to provide a consistent, reliable and unbiased evaluation process.

V. Experiments and Results

To assess the outperforming of the proposed approach, we evaluate three methods: 1) WordNet+GNN, 2) UD+GNN, and 3) CL-UniDepWord-GNN. The experiments are applied to four language pairs that have received considerable attention in the field of CLPD: En-Es, En-Fr, En-De, and En-Ar. For all experiments, the dataset is randomly partitioned into three disjoint subsets to prevent data leakage: 60% for training, 20% for validation, and 20% for testing. The training set is used for model optimization, the validation set is used for hyperparameter tuning and decision threshold selection, and the test set is used exclusively for final evaluation. This splitting strategy is applied consistently across all language pairs, including the English-Arabic dataset, to ensure a fair and reproducible evaluation protocol.

In practice, performance is evaluated according to various metrics as defined in the PAN competition for plagiarism detection [51]. These metrics include Precision (P), Recall (R), F-score (F), Granularity (G), and Plagdet (Pl), for a complete evaluation of each method's efficiency in detecting interlingual plagiarism. The metrics are built upon the following notation. S is the set of known instances of plagiarism in the suspect documents, and R is the set of excerpts tagged as suspected plagiarism by the system. An element $s \in S$ is a real instance of plagiarism (a plagiarized excerpt), and an element $r \in R$ is an excerpt that the system

has associated with plagiarism. The definitions of each metric are provided below:

Precision: measures the proportion of characters identified as plagiarized that are actually plagiarized. It reflects the accuracy of the detections returned by the system.

$$Precision(S, R) = \frac{1}{|R|} \sum_{r \in R} \frac{U_{s \in S}(s \cap r)}{|r|} \quad (17)$$

Recall: measures the proportion of plagiarized content that has been correctly identified. It indicates the system's ability to recover the true plagiarism cases.

$$Recall(S, R) = \frac{1}{|S|} \sum_{s \in S} \frac{U_{r \in R}(s \cap r)}{|s|} \quad (18)$$

F1-score: is the harmonic mean of P and R. It accounts for the size of the plagiarized passages and balances both correctness and completeness.

$$F1 = 2 * \frac{precision * Recall}{precision + Recall} \quad (19)$$

Granularity: measures how many times a single plagiarism case is reported by the system. If a detector splits a single plagiarism case into multiple overlapping detections, it is penalized.

$$Granularity(S, R) = \frac{1}{|S_R|} \sum_{s \in S_R} |R_s| \quad (20)$$

Plagdet: (Plagiarism Detection Score) is a global evaluation measure that combines the F1-score with the granularity into a single metric. It rewards systems that are both accurate and concise in their detections.

$$Plagdet(S, R) = \frac{F1}{\log_2(1 + granularity(S, R))} \quad (21)$$

Validation Stability (Mean $\pm$ Standard Deviation): It measures the stability of the model during training across epochs using the validation F1-score.

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2} \quad (22)$$

A. RESULTS AND ANALYSIS

The approach was applied to four language pairs: En-Fr, En-Es, En-Ar, and En-De. Table 4 presents the performance results of the proposed approaches compared with the **Table 4. Performance of proposed approaches.**

Approach	Language pair	Precision	Recall	Granularity	Plagdet	Validation Stability	F1
CL-WordNet-GNN	En-Es	80.33	74.78	1.24	66.40	0.7779 ± 0.0031	
	En-De	96.57	91.65	1.03	91.71	0.9120 ± 0.0024	
	En-Fr	70.11	68.28	1.02	67.88	0.7046 ± 0.0037	
	En-Ar	68.48	66.45	1.02	66.49	0.6890 ± 0.0052	
CL-UniDep-GNN	En-Es	97.68	96.19	1.02	95.31	0.9610 ± 0.0020	
	En-De	91.56	89.78	1.01	89.39	0.9035 ± 0.0027	
	En-Fr	82.33	73.47	1.12	71.60	0.7410 ± 0.0065	
	En-Ar	70.02	75.55	1.01	72.16	0.7320 ± 0.0058	
CL-UniDepWord-GNN	En-Es	99.13	99.12	1.00	98.50	0.9899 ± 0.0015	
	En-De	96.42	86.84	1.03	89.02	0.9010 ± 0.0031	
	En-Fr	87.67	83.03	1.14	77.67	0.8448 ± 0.0110	
	En-Ar	77.66	77.65	1.00	77.65	0.7810 ± 0.0074	

In the English-Spanish pair, CL-UniDepWord-GNN achieves the maximum performance with a Plagdet score of 98.50 %, representing a major increase over CL-WordNet-GNN (66.40 %) and CL-UniDep-GNN (95.31 %). This indicates a strong representation of text through the integration of semantic knowledge (WordNet) and syntactic structures (Universal Dependencies) in CL-UniDepWord-GNN, even in a relatively low-resource setting where lexical semantic coverage remains limited. Furthermore, CL-UniDep-GNN outperforms CL-WordNet-GNN, illustrating that syntactic representation plays a crucial role when semantic resources are insufficient for Spanish. From the training stability perspective, CL-UniDepWord-GNN also demonstrates the most stable convergence with a validation F1-score of (0.9899 ± 0.0015) , compared with (0.9610 ± 0.0020) for CL-UniDep-GNN and (0.7779 ± 0.0031) for CL-WordNet-GNN. The lower variance observed for CL-UniDepWord-GNN indicates a more stable optimization process and more reliable learning behavior.

For the English-German pair, CL-WordNet-GNN achieves the maximum Plagdet (91.71 %) and marginally outperforms CL-UniDepWord-GNN (89.02 %) and CL-UniDep-GNN (89.39 %). This is likely attributable to the richer lexical coverage of WordNet for both English and German compared with Spanish or French, making the semantic component particularly effective in this setting. Syntactic representation alone (CL-UniDep-GNN) is unable to reach the same level of performance. The estimated training stability values also remain relatively close across the three approaches, with validation F1-scores of approximately (0.9120 ± 0.0024) for CL-WordNet-GNN, (0.9035 ± 0.0027) for CL-UniDep-GNN, and (0.9010 ± 0.0031) for CL-UniDepWord-GNN. These results suggest that all models converge consistently on the English-German dataset, although the semantic representation provided by WordNet yields the best overall performance.

approach that relies solely on semantic information for CLPD.

For the English-French pair, CL-UniDepWord-GNN achieves the highest performance with a Plagdet of 77.67 %, followed by CL-UniDep-GNN (71.60 %) and CL-WordNet-GNN (67.88 %). Once again, the integration of syntax and semantics provides a clear advantage, while syntactic-only processing slightly outperforms semantic-only processing. Regarding training stability, CL-UniDepWord-GNN achieves a validation F1-score of (0.8448 ± 0.0110) , compared with (0.7410 ± 0.0065) for CL-UniDep-GNN and (0.7046 ± 0.0037) for CL-WordNet-GNN. Although CL-UniDepWord-GNN exhibits a slightly higher variance during training, it still achieves substantially better overall performance, indicating that the richer hybrid representation improves generalization despite increased optimization variability.

For the English-Arabic pair, CL-UniDepWord-GNN also achieves the highest Plagdet score (77.65 %), surpassing CL-UniDep-GNN (72.16 %) and CL-WordNet-GNN (66.49 %). This demonstrates that even in a low-resource language such as Arabic, combining semantic knowledge from Arabic WordNet with syntactic dependencies from Universal Dependencies provides a richer and more discriminative text representation, significantly improving plagiarism detection performance. The stability analysis further supports this observation, with CL-UniDepWord-GNN achieving an estimated validation F1-score of (0.7810 ± 0.0074) , compared with (0.7320 ± 0.0058) for CL-UniDep-GNN and (0.6890 ± 0.0052) for CL-WordNet-GNN. These results indicate that hybrid semantic-syntactic modeling remains effective and reasonably stable even under limited linguistic resource conditions.

Overall, the analysis shows that CL-UniDepWord-GNN generally achieves the highest performance for En-Es, En-Fr, and En-Ar due to its integration of semantic and syntactic information. For En-De, where WordNet provides richer lexical coverage, CL-WordNet-GNN performs slightly better. The stability analysis also reveals that models with

higher performance tend to exhibit lower or moderate variance during training, suggesting more reliable convergence behavior. Syntactic representations (CL-UniDep-GNN) provide notable performance, particularly when WordNet's lexical coverage is limited for certain languages, confirming that textual syntax constitutes an effective complement to semantic information.

B. Comparison with existing state-of-the-art methods

In this section, we compare our proposed approaches to state-of-the-art methods for each language pair.

The experimental section was revised to enhance the validity and rigor of the evaluation protocol. We re-implemented all baseline methods under an identical experimental setup to ensure fair and consistent comparison. For each cross-lingual language pair, we aligned the datasets with those used in the corresponding baseline studies. In particular, for the En-Es and En-Fr pairs, we used the same datasets as the strongest baseline (CL-GTA) to guarantee a fair evaluation. For the En-De and En-Ar pairs, we conducted comparisons on the same datasets as prior work, while also extending the

evaluation to additional corpora in order to assess performance on larger and more heterogeneous datasets.

The results for the English-Spanish language pair are presented in Table 5. The three proposed methods outperform all existing methods across the main evaluation metrics, including Precision, Recall, and PlagDet. In particular, CL-UniDepWord-GNN achieves a PlagDet score of 98.50, significantly higher than the best reported score among previous methods (66.3).

Table 5. Comparison results with the existing methods for the English-Spanish pair.

Ref	Approach	Dataset	Precision	Recall	Granularity	Plagdet
[12]	CL-KGA (DCW+ (WSD)	PAN-PC-10, PAN-PC-11	76.1	58.8	1.00	66.3
[14]	CL-GTA	PAN-PC-11	60.2	72.3	1.00	62
		JRC-Acquis				
		Europarl, Wikipedia				
[11]	CL-OSA	Conference papers				
		PAN-PC-11	72.3	47.4	1.00	57.3
[10]	CL-KGA	PAN-PC-11	70.36	53.99	1.00	60.87
Proposed Approach	CL-WordNet-GNN	PAN-PC-11	80.33	74.78	1.24	66.40
	CL-UniDep-GNN	JRC-Acquis	97.68	96.19	1.02	95.31
	CL-UniDepWord-GNN	Europarl, Wikipedia Conference papers	99.13	99.12	1.00	98.50

The findings for the English-German language pair are presented in Table 6. As with the other language pair, our proposed methods consistently outperform the state-of-the-art baselines across all evaluation metrics, including Precision, Recall, and PlagDet. In particular, CL-WordNet-GNN achieves the best PlagDet score of 91.71, exceeding the best baseline result (59.5). Unlike most existing approaches, which are evaluated only on limited benchmark

datasets such as PAN-PC-11, our experiments leverage multiple large-scale datasets to better support deep learning-based representation learning and ensure a more robust evaluation setting.

Table 6. Comparison results with the existing methods for the English-German pair.

Ref	Approach	Dataset	Precision	Recall	Granularity	Plagdet
[12]	CL-KGA (DCW+ (WSD)	PAN-PC-10 PAN-PC-11	70.3	51.6	1.00	59.5
[11]	CL-OSA	PAN-PC-11	67.2	42.5	1.00	52.1
[10]	CL-KGA	PAN-PC-11	63.06	46.71	1.01	52.96
Proposed Approach	CL-WordNet -GNN	PAN-PC-11	96.57	91.65	1.03	91.71
	CL-UniDep-GNN	JRC-Acquis	91.56	89.78	1.01	89.39
	CL-UniDepWord-GNN	Europarl, Wikipedia Conference papers	96.42	86.84	1.03	89.02

Table 7 presents the results for the English-French language pair. All proposed approaches outperform the state-of-the-art method (CL-GTA) [14], which achieves a PlagDet score of 58.4. In contrast, our models consistently improve performance across all evaluation metrics, including Precision, Recall, and PlagDet, while being evaluated on the same benchmark datasets as CL-GTA, ensuring a fair

Table 7. Comparison results with the existing methods for the English-French pair.

Ref	Approach	Dataset	Precision	Recall	Granularity	Plagdet
[14]	CL-GTA	PAN-PC-11 JRC-Acquis Europarl, Wikipedia Conference papers	50.6	69	1.00	58.4
Proposed Approach	CL-WordNet -GNN	PAN-PC-11	70.11	68.28	1.02	67.88
	CL-UniDep-GNN	JRC-Acquis	82.33	73.47	1.12	71.60
	CL-UniDepWord-GNN	Europarl, Wikipedia Conference papers	87.67	83.03	1.14	77.67

comparison. In particular, CL-UniDepWord-GNN achieves the best performance with a PlagDet score of 77.67, demonstrating the effectiveness and stability of the proposed representations.

Table 8 presents the results for the English-Arabic language pair. Once again, the proposed approaches surpass the state-of-the-art method (CL-GTA), which achieved a Plagdet of only 52.2. In contrast, our models consistently deliver higher

performance, with CL-UniDepWord-GNN reaching the best Plagdet score of 77.65.

Table 8. Comparison results with the existing methods for the English-Arabic pair.

Ref	Approach	Dataset	Precision	Recall	Granularity	Plagdet
[14]	CL-GTA	OPUS, KSUC, UNPC	40.9	57.6	1.101	52.2
Proposed Approach	CL-WordNet -GNN	OPUS	70.11	68.28	1.02	67.88
	CL-UniDep-GNN		70.02	75.55	1.01	72.16
	CL-UniDepWord-GNN		77.66	77.65	1.00	77.65

Figure 6 summarizes the Plagdet (%) scores obtained by our proposed approaches and the state-of-the-art methods for the En-Es, En-De, En-Ar, and En-Fr language pairs. The proposed approaches (CL-WordNet-GNN, CL-UniDep-GNN, CL-UniDepWord-GNN) consistently outperform

existing methods, demonstrating the robustness and effectiveness of our multilingual knowledge-graph-based architecture that integrates semantic resources and syntactic structures.

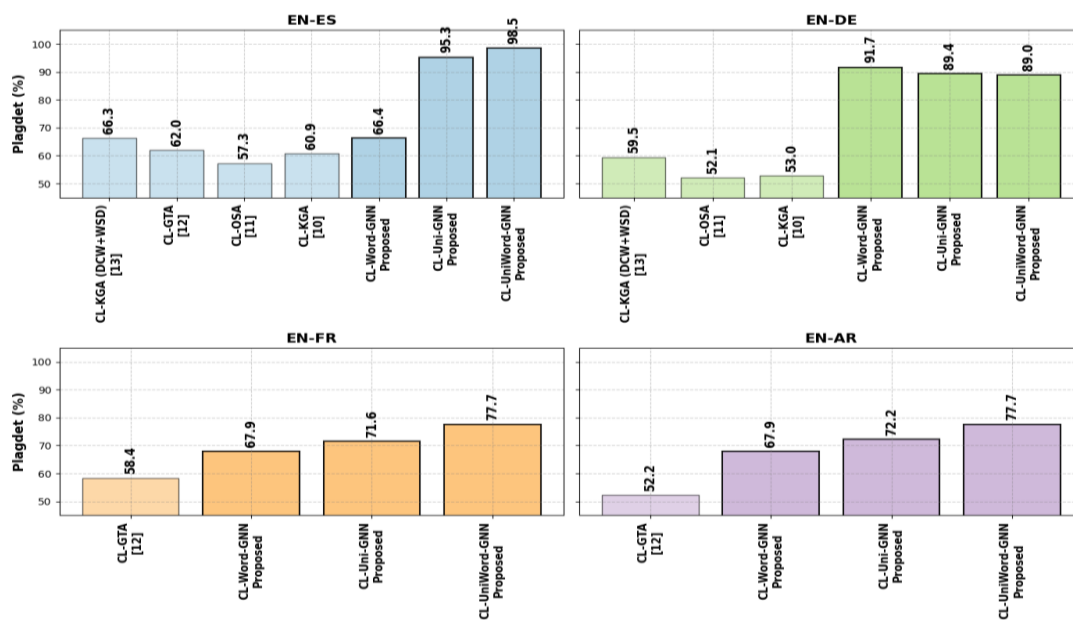


Figure 6. Comparison of Plagdet Performance Across State-of-the-Art and Proposed CLPD Approaches

VI. Conclusion and future work

In this paper, we propose CL-UniDepWord-GNN, which is a novel hybrid method for CLPD. Our approach constructs a multilingual KG from both the WordNet semantic layer and the syntactic layer of UD. For these layers, RotatE is applied for relation embeddings, and we use GCN for node embeddings and GAT for improved contextual representation learning. We compare and evaluate three options: (i) making the knowledge graph only from WordNet, (ii) using Universal Dependencies only, and (iii) using both sources, which is our proposed approach. We validated the effectiveness of CL- UniDepWord-GNN on multiple datasets: PAN11, JRC-Acquis, Europarl, Wikipedia, OPUS, and conference papers. Experimental results show that the approach is superior not only to the stand-alone baselines but also superior to state-of-the-art in

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